ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 3.01: Diophantine Equations, SFFT, Bezout & Floor Functions

Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 10x + 38 in the domain of Diophantine Equations, SFFT & Floor Functions. By completing the square f_1x = x - h^2 + k , find the sum k + f_17 .

[ ] A: 35
[ ] B: 17
[ ] C: 30
[ ] D: 28

Q2. A mathematical model in Diophantine Equations, SFFT & Floor Functions is defined by the cubic function y_2x = 4x^3 - 5x^2 + 15x - 10 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 123
[ ] B: 78
[ ] C: 41
[ ] D: 93

Q3. In Diophantine Equations, SFFT & Floor Functions, two displacement vectors are given by mathbf{u}_{3} = langle 5, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 78
[ ] B: -11
[ ] C: 89
[ ] D: 190

Q4. The linear growth profile shown in the diagram for Diophantine Equations, SFFT & Floor Functions follows an arithmetic progression with first term a_1 = 6 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 90
[ ] B: 720
[ ] C: 810
[ ] D: 1350

Q5. In Diophantine Equations, SFFT & Floor Functions, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 20x,mathrm{d}x .

[ ] A: 112
[ ] B: 108
[ ] C: 88
[ ] D: 45

Q6. In Diophantine Equations, SFFT & Floor Functions, consider the quadratic function f_6x = x^2 - 12x + 121 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_613 .

[ ] A: 225
[ ] B: 134
[ ] C: 212
[ ] D: 219

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Q7. The graph illustrates the cubic model y_7x = 4x^3 - 6x^2 + 33x - 17 studied in Diophantine Equations, SFFT & Floor Functions. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 705
[ ] B: 801
[ ] C: 672
[ ] D: 117

Q8. In Diophantine Equations, SFFT & Floor Functions, two displacement vectors are given by mathbf{u}_{8} = langle 6, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 13
[ ] B: 98
[ ] C: 85
[ ] D: 168